

Mintuu AI Ecosystem Complete Technical Snapshot

Generated from local workspace: `/Users/mridulsoliwal/Documents/All_Projects/AI_Agents/mintuu_ai_ecosystem`
Generation date: 2026-05-28 Asia/Kolkata.
Purpose: brief an external AI assistant for strategic planning, product direction, feature planning, and integration prompts.

1. Project Identity

Field	Value
Repository folder	<code>AI_Agents/mintuu_ai_ecosystem</code>
Package name	<code>mintuu-ai-ecosystem</code> from <code>pyproject.toml</code>
Package version	<code>1.0.0</code> from <code>pyproject.toml</code>
Runtime/settings version	<code>3.0.0</code> from <code>config/settings.py</code>
FastAPI app title	<code>Mintuu AI Ecosystem v2</code> from <code>api/app.py</code>
README title	<code>Mintuu AI Ecosystem v3</code>
One-sentence description	An autonomous AI-native business operating system that coordinates specialized departmental AI agents through a central orchestration layer, workflow engine, memory system, tools, and a real-time dashboard.
Problem solved	It turns isolated AI agents into a coordinated digital workforce that can route work, execute multi-step workflows, remember results, collaborate, expose APIs, and present state through a dashboard.
Built for	Founders, builders, AI-agent developers, small teams, and operators who want a local/prototype autonomous business operations platform.

Important identity note: the project has version naming drift. Packaging says `1.0.0`, settings say `3.0.0`, FastAPI title says `v2`, and README says `v3`. External planning should treat this as a v3 prototype with some v2 naming left in code.

2. Complete Tech Stack

Technology	Exact observed version	Role	Files using it
Python	<code>Python 3.14.4</code>	Backend language and runtime	All <code>.py</code> files
FastAPI	<code>0.136.1</code>	HTTP API, page routes, WebSocket/webhook routers	<code>api/app.py</code> , <code>api/auth_routes.py</code> , <code>api/webhooks.py</code> , <code>api/websocket.py</code>
Uvicorn	<code>0.46.0</code>	ASGI dev/production server	README, Makefile, Render start command
Pydantic	<code>2.13.4</code>	Request models and settings models	<code>api/app.py</code> , <code>api/auth_routes.py</code> , <code>config/settings.py</code> , <code>core/events/event_types.py</code>
Jinja2	<code>3.1.6</code>	Declared template dependency; templates are currently read directly as files	<code>pyproject.toml</code> ; dashboard templates
SQLite	<code>stdlib sqlite3 plus .db files</code>	Primary persistent relational storage	<code>database/models.py</code> , <code>database/mintuu_ecosystem.db</code>
ChromaDB	<code>1.5.9</code>	Persistent vector store for semantic organizational memory	<code>core/memory/vector_memory.py</code> , <code>database/chroma_db/</code>
OpenAI SDK	<code>2.36.0</code>	OpenAI LLM provider adapter	<code>core/llm/providers/openai_provider.py</code>
Anthropic SDK / HTTP API	<code>0.100.0</code>	Declared/installed; current Anthropic provider is mocked and uses <code>httpx</code> import	<code>core/llm/providers/anthropic_provider.py</code>
Ollama HTTP API	<code>local service, model configured by env</code>	Local LLM provider	<code>core/llm/providers/ollama_provider.py</code>

Technology	Exact observed version	Role	Files using it
Groq API	external API; package not required	Cloud/free-tier LLM provider via aiohttp	core/llm/providers/groq_provider.py
aiohttp	3.13.5	Async HTTP client for Ollama/Groq and test script	core/llm/providers/ollama_provider.py, core/llm/providers/groq_provider.py, test_ollama.py
httpx	0.28.1	Imported by Anthropic provider	core/llm/providers/anthropic_provider.py
bcrypt	5.0.0	Password hashing and verification	api/auth.py
python-jose	3.5.0	JWT encode/decode	api/auth.py
python-multipart	0.0.27	FastAPI form/file upload support dependency	pyproject.toml
Redis	7.4.0	Declared production dependency and Docker service for Celery broker	docker-compose.yml, core/distributed/celery_app.py
Celery	5.6.3	Distributed worker task queue	core/distributed/celery_app.py, core/distributed/tasks.py, docker-compose.yml
Flower	Docker image mher/flower`	Celery monitoring UI	docker-compose.yml
PyGithub	2.9.1	GitHub automation dependency; standalone implementation is partial	requirements_v3.txt, tools/implementations/github_tool.py
Playwright	1.59.0	Browser automation dependency; standalone implementation is partial	requirements_v3.txt, tools/implementations/browser_tool.py
python-dotenv	1.2.2	Loads .env at startup	config/settings.py
HTML/CSS/JavaScript	browser-native	Dashboard, landing page, auth pages	dashboard/templates/*, dashboard/static/*
Alpine.js	CDN 3.x.x`	Reactive dashboard controller	dashboard/templates/dashboard.html, dashboard/static/js/app.js
marked.js	CDN latest from marked/marked.min.js`	Markdown rendering in dashboard if needed	dashboard/templates/dashboard.html
Three.js	CDN r128	Landing hero particle network	dashboard/static/js/hero3d.js
Docker Compose	file format 3.8	Local multi-service orchestration	docker-compose.yml
Render	free web service config	Backend deployment	render.yaml
Vercel	static/proxy config	Frontend deployment attempt	vercel.json
Kubernetes	apps/v1 Deployment	Production deployment sketch	deployment/kubernetes/deployment.yaml
Nginx	reverse proxy config	Load balancing and WebSocket proxying sketch	deployment/nginx/nginx.conf
Terraform	AWS provider sketch	EKS infrastructure sketch	deployment/terraform/main.tf
pytest	9.0.3	Test runner; no tests currently collected	Makefile, tests/
ruff	0.15.12	Declared dev linter	pyproject.toml

3. Complete File Structure

```

.
├── .DS_Store
├── .env
├── .env.example
├── .file_inventory.txt
├── .pytest_cache/
│   ├── .gitignore
│   ├── CACHEDIR.TAG
│   ├── README.md
│   └── v/
│       └── cache/
│           └── nodeids
├── .section5_generated.md
├── Makefile
├── README.md
├── __init__.py
├── __pycache__/
│   ├── __init__.cpython-314.pyc
│   └── run_flagships_v2.cpython-314.pyc
├── agents/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   └── analytics_agent/
│       ├── __init__.py
│       └── agent.py
│   ├── base_agent/
│       ├── __init__.py
│       ├── __pycache__/
│       │   ├── __init__.cpython-314.pyc
│       │   └── base.cpython-314.pyc
│       └── base.py
│   ├── ceo_agent/
│       ├── __init__.py
│       ├── __pycache__/
│       │   ├── __init__.cpython-314.pyc
│       │   ├── agent.cpython-314.pyc
│       │   └── system_prompt.cpython-314.pyc
│       ├── agent.py
│       └── system_prompt.py
│   ├── finance_agent/
│       ├── __init__.py
│       ├── __pycache__/
│       │   ├── __init__.cpython-314.pyc
│       │   ├── agent.cpython-314.pyc
│       │   └── system_prompt.cpython-314.pyc
│       ├── agent.py
│       └── system_prompt.py
│   ├── hr_agent/
│       ├── __init__.py
│       ├── __pycache__/
│       │   ├── __init__.cpython-314.pyc
│       │   └── agent.cpython-314.pyc
│       └── agent.py
│   ├── infrastructure_agent/
│       ├── __pycache__/
│       │   └── agent.cpython-314.pyc
│       └── agent.py
│   ├── legal_agent/
│       ├── __init__.py
│       └── agent.py
│   ├── marketing_agent/
│       ├── __init__.py
│       ├── __pycache__/
│       │   ├── __init__.cpython-314.pyc
│       │   ├── agent.cpython-314.pyc
│       │   └── system_prompt.cpython-314.pyc
│       ├── agent.py
│       └── system_prompt.py
│   ├── operations_agent/
│       ├── __init__.py
│       ├── __pycache__/
│       │   ├── __init__.cpython-314.pyc
│       │   └── agent.cpython-314.pyc

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```

├── agent.py
├── product_strategy_agent/
├── production_agent/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── agent.cpython-314.pyc
│   └── agent.py
├── research_agent/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   ├── agent.cpython-314.pyc
│   │   └── system_prompt.cpython-314.pyc
│   ├── agent.py
│   └── system_prompt.py
├── security_agent/
│   ├── __pycache__/
│   │   └── agent.cpython-314.pyc
│   └── agent.py
├── support_agent/
├── api/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   ├── app.cpython-314.pyc
│   │   ├── auth.cpython-314.pyc
│   │   ├── auth_routes.cpython-314.pyc
│   │   ├── webhooks.cpython-314.pyc
│   │   └── websocket.cpython-314.pyc
│   ├── app.py
│   ├── auth.py
│   ├── auth_routes.py
│   ├── webhooks.py
│   └── websocket.py
├── automation/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── autonomous_engine.cpython-314.pyc
│   └── autonomous_engine.py
├── build_report_workflow.py
├── cli/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── main.cpython-314.pyc
│   └── main.py
├── config/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── settings.cpython-314.pyc
│   └── settings.py
├── core/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   └── adaptation/
│       ├── behavior_optimizer.py
│       ├── feedback_processor.py
│       ├── learning_engine.py
│       ├── strategy_evaluator.py
│       └── workflow_optimizer.py
├── approval/
│   ├── approval_engine.py
│   ├── escalation_manager.py
│   ├── human_review.py
│   └── risk_analyzer.py
├── communication/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── message_bus.cpython-314.pyc
│   └── message_bus.py

```

```

└─ distributed/
  └─ __init__.py
  └─ celery_app.py
  └─ tasks.py
└─ events/
  └─ __init__.py
  └─ __pycache__/
    └─ __init__.cpython-314.pyc
    └─ event_bus.cpython-314.pyc
    └─ event_types.cpython-314.pyc
  └─ event_bus.py
  └─ event_types.py
└─ goals/
  └─ __init__.py
  └─ goal_engine.py
└─ llm/
  └─ __init__.py
  └─ __pycache__/
    └─ __init__.cpython-314.pyc
    └─ llm_manager.cpython-314.pyc
    └─ provider_router.cpython-314.pyc
    └─ reasoning_engine.cpython-314.pyc
    └─ token_tracker.cpython-314.pyc
  └─ context_builder.py
  └─ llm_manager.py
  └─ provider_router.py
  └─ providers/
    └─ __init__.py
    └─ __pycache__/
      └─ __init__.cpython-314.pyc
      └─ anthropic_provider.cpython-314.pyc
      └─ base_provider.cpython-314.pyc
      └─ groq_provider.cpython-314.pyc
      └─ ollama_provider.cpython-314.pyc
      └─ openai_provider.cpython-314.pyc
    └─ anthropic_provider.py
    └─ base_provider.py
    └─ groq_provider.py
    └─ ollama_provider.py
    └─ openai_provider.py
  └─ reasoning_engine.py
  └─ token_tracker.py
└─ memory/
  └─ __init__.py
  └─ __pycache__/
    └─ __init__.cpython-314.pyc
    └─ memory_manager.cpython-314.pyc
    └─ vector_memory.cpython-314.pyc
  └─ advanced/
    └─ long_term_knowledge.py
    └─ memory_ranker.py
    └─ memory_summarizer.py
    └─ organizational_brain.py
    └─ semantic_learning.py
  └─ memory_manager.py
  └─ vector_memory.py
└─ multimodal/
  └─ document_parser.py
  └─ multimodal_router.py
  └─ screenshot_reasoner.py
  └─ vision_engine.py
└─ orchestration/
  └─ __init__.py
  └─ __pycache__/
    └─ __init__.cpython-314.pyc
    └─ agent_registry.cpython-314.pyc
    └─ collaboration_engine.cpython-314.pyc
    └─ orchestration_manager.cpython-314.pyc
    └─ task_router.cpython-314.pyc
  └─ agent_registry.py
  └─ collaboration_engine.py
  └─ orchestration_manager.py
  └─ task_router.py
└─ planning/
  └─ adaptive_planner.py

```

```

├── dependency_graph.py
├── execution_forecaster.py
├── strategic_planner.py
├── task_decomposer.py
├── reflection/
│   └── reflection_engine.py
├── state/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── state_manager.cpython-314.pyc
│   └── state_manager.py
├── workflows/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   ├── flagship_workflows.cpython-314.pyc
│   │   └── workflow_engine.cpython-314.pyc
│   ├── flagship_workflows.py
│   └── workflow_engine.py
├── dashboard/
│   ├── __init__.py
│   ├── static/
│   │   ├── assets/
│   │   ├── css/
│   │   │   ├── app.css
│   │   │   ├── base.css
│   │   │   ├── dashboard.css
│   │   │   ├── landing.css
│   │   │   └── theme.css
│   │   └── js/
│   │       ├── app.js
│   │       ├── auth.js
│   │       ├── dashboard.js
│   │       └── hero3d.js
│   ├── templates/
│   │   ├── dashboard.html
│   │   ├── landing.html
│   │   ├── login.html
│   │   ├── logo.html
│   │   └── signup.html
├── data/
│   └── outputs/
├── database/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   └── models.cpython-314.pyc
│   ├── chroma_db/
│   │   ├── 909af621-9653-4797-be07-b16f020b8c87/
│   │   │   ├── data_level0.bin
│   │   │   ├── header.bin
│   │   │   ├── length.bin
│   │   │   └── link_lists.bin
│   │   └── chroma.sqlite3
│   ├── mintuu_ecosystem.db
│   └── models.py
├── deployment/
│   ├── kubernetes/
│   │   └── deployment.yaml
│   ├── nginx/
│   │   └── nginx.conf
│   ├── production/
│   ├── terraform/
│   │   └── main.tf
├── docker-compose.yml
├── docs/
├── execution_trace.json
├── generate_architecture.py
├── integrations/
│   └── __init__.py
├── logs/
│   └── workflow_1_execution.log
├── mintuu_ai_ecosystem.egg-info/
│   └── PKG-INFO

```

```
├── SOURCES.txt
├── dependency_links.txt
├── entry_points.txt
├── requires.txt
├── top_level.txt
├── mintuu_ecosystem.db
├── mintuu_flagship_report.html
├── mintuu_master_doc.md
├── mintuu_master_doc.pdf
├── mintuu_master_doc_final.md
├── mintuu_master_doc_final.pdf
├── monitoring/
│   ├── orchestration_metrics.py
│   ├── performance_monitor.py
│   ├── system_profiler.py
│   ├── tracing_engine.py
│   └── workflow_analytics.py
├── pyproject.toml
├── raw_reasoning_log.txt
├── render.yaml
├── report/
│   ├── .DS_Store
│   ├── output/
│   │   ├── mintuu_flagship_report.html
│   │   ├── mintuu_flagship_report.md
│   │   ├── mintuu_flagship_report.pdf
│   │   └── screenshots/
│   │       ├── chat_workflow.png
│   │       ├── dashboard_bottom.png
│   │       ├── dashboard_overview.png
│   │       ├── dashboard_top.png
│   │       ├── final_state.png
│   │       └── workflow_completed.png
│   └── w1_dashboard.png
├── requirements_v3.txt
├── run_flagships.py
├── run_flagships_v2.py
├── test_ollama.py
├── test_reasoning.py
├── test_workflow.py
├── tests/
│   └── __init__.py
├── tools/
│   ├── __init__.py
│   ├── __pycache__/
│   │   ├── __init__.cpython-314.pyc
│   │   ├── execution_manager.cpython-314.pyc
│   │   └── tool_registry.cpython-314.pyc
│   ├── execution_manager.py
│   ├── implementations/
│   │   ├── __init__.py
│   │   ├── browser_tool.py
│   │   ├── email_tool.py
│   │   └── github_tool.py
│   └── tool_registry.py
└── vercel.json
```

File-by-File Purpose

File	Purpose
.DS_Store	macOS Finder metadata file; not part of application logic.
.env	Local environment file containing API keys and runtime settings; values are intentionally not reproduced in this snapshot.
.env.example	Template showing expected environment variables for local and deployed environments.
.file_inventory.txt	Plain text inventory, generated section, or raw reasoning log artifact.
.pytest_cache/.gitignore	Project file or artifact used by the repository.

File	Purpose
<code>.pytest_cache/CACHEDIR.TAG</code>	Project file or artifact used by the repository.
<code>.pytest_cache/README.md</code>	Markdown documentation or generated report artifact.
<code>.pytest_cache/v/cache/nodeids</code>	Project file or artifact used by the repository.
<code>.section5_generated.md</code>	Markdown documentation or generated report artifact.
<code>Makefile</code>	Convenience commands for installing, running the dev server, testing, and starting an ngrok tunnel.
<code>README.md</code>	Top-level project overview, architecture summary, stack notes, and basic run commands.
<code>__init__.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>__pycache__/run_flagships_v2.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/__init__.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>agents/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/analytics_agent/__init__.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>agents/analytics_agent/agent.py</code>	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
<code>agents/base_agent/__init__.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>agents/base_agent/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/base_agent/__pycache__/base.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/base_agent/base.py</code>	Abstract base class that implements shared agent lifecycle, messaging, tool use, memory use, logging, and task handling.
<code>agents/ceo_agent/__init__.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>agents/ceo_agent/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/ceo_agent/__pycache__/agent.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/ceo_agent/__pycache__/system_prompt.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>agents/ceo_agent/agent.py</code>	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
<code>agents/ceo_agent/system_prompt.py</code>	Defines the system prompt used by the LLM reasoning layer for that agent.
<code>agents/finance_agent/__init__.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>agents/finance_agent/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.

File	Purpose
agents/finance_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/finance_agent/__pycache__/system_prompt.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/finance_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/finance_agent/system_prompt.py	Defines the system prompt used by the LLM reasoning layer for that agent.
agents/hr_agent/__init__.py	Python source module for project logic, scripts, tests, or utilities.
agents/hr_agent/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/hr_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/hr_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/infrastructure_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/infrastructure_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/legal_agent/__init__.py	Python source module for project logic, scripts, tests, or utilities.
agents/legal_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/marketing_agent/__init__.py	Python source module for project logic, scripts, tests, or utilities.
agents/marketing_agent/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/marketing_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/marketing_agent/__pycache__/system_prompt.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/marketing_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/marketing_agent/system_prompt.py	Defines the system prompt used by the LLM reasoning layer for that agent.
agents/operations_agent/__init__.py	Python source module for project logic, scripts, tests, or utilities.
agents/operations_agent/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/operations_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/operations_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/production_agent/__init__.py	Python source module for project logic, scripts, tests, or utilities.

File	Purpose
agents/production_agent/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/production_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/production_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/research_agent/__init__.py	Python source module for project logic, scripts, tests, or utilities.
agents/research_agent/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/research_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/research_agent/__pycache__/system_prompt.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/research_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
agents/research_agent/system_prompt.py	Defines the system prompt used by the LLM reasoning layer for that agent.
agents/security_agent/__pycache__/agent.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
agents/security_agent/agent.py	Defines a specialized Mintuu agent class, its identity, capabilities, planning logic, execution logic, and result summary behavior.
api/__init__.py	FastAPI API module for pages, REST endpoints, auth routes, webhooks, or WebSocket updates.
api/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
api/__pycache__/app.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
api/__pycache__/auth.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
api/__pycache__/auth_routes.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
api/__pycache__/webhooks.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
api/__pycache__/websocket.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
api/app.py	FastAPI API module for pages, REST endpoints, auth routes, webhooks, or WebSocket updates.
api/auth.py	FastAPI API module for pages, REST endpoints, auth routes, webhooks, or WebSocket updates.
api/auth_routes.py	FastAPI API module for pages, REST endpoints, auth routes, webhooks, or WebSocket updates.
api/webhooks.py	FastAPI API module for pages, REST endpoints, auth routes, webhooks, or WebSocket updates.
api/websocket.py	FastAPI API module for pages, REST endpoints, auth routes, webhooks, or WebSocket updates.
automation/__init__.py	Autonomous recurring task engine and package marker.

File	Purpose
automation/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
automation/__pycache__/autonomous_engine.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
automation/autonomous_engine.py	Autonomous recurring task engine and package marker.
build_report_workflow.py	Python source module for project logic, scripts, tests, or utilities.
cli/__init__.py	Command-line package module and entry point placeholder for the mintuu console script.
cli/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
cli/__pycache__/main.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
cli/main.py	Command-line package module and entry point placeholder for the mintuu console script.
config/__init__.py	Centralized application settings loaded from environment variables and defaults.
config/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
config/__pycache__/settings.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
config/settings.py	Centralized application settings loaded from environment variables and defaults.
core/__init__.py	Python source module for project logic, scripts, tests, or utilities.
core/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/adaptation/behavior_optimizer.py	Placeholder or early-stage adaptation/learning component for future optimization features.
core/adaptation/feedback_processor.py	Placeholder or early-stage adaptation/learning component for future optimization features.
core/adaptation/learning_engine.py	Placeholder or early-stage adaptation/learning component for future optimization features.
core/adaptation/strategy_evaluator.py	Placeholder or early-stage adaptation/learning component for future optimization features.
core/adaptation/workflow_optimizer.py	Placeholder or early-stage adaptation/learning component for future optimization features.
core/approval/approval_engine.py	Approval, escalation, human review, or risk-analysis component, currently mostly skeletal.
core/approval/escalation_manager.py	Approval, escalation, human review, or risk-analysis component, currently mostly skeletal.
core/approval/human_review.py	Approval, escalation, human review, or risk-analysis component, currently mostly skeletal.
core/approval/risk_analyzer.py	Approval, escalation, human review, or risk-analysis component, currently mostly skeletal.
core/communication/__init__.py	Message bus implementation for agent-to-agent and orchestrator-to-agent communication.
core/communication/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.

File	Purpose
core/communication/__pycache__/message_bus.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/communication/message_bus.py	Message bus implementation for agent-to-agent and orchestrator-to-agent communication.
core/distributed/__init__.py	Celery distributed execution configuration and background task definitions.
core/distributed/celery_app.py	Celery distributed execution configuration and background task definitions.
core/distributed/tasks.py	Celery distributed execution configuration and background task definitions.
core/events/__init__.py	Async event bus and event type definitions used by webhook/event-driven flows.
core/events/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/events/__pycache__/event_bus.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/events/__pycache__/event_types.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/events/event_bus.py	Async event bus and event type definitions used by webhook/event-driven flows.
core/events/event_types.py	Async event bus and event type definitions used by webhook/event-driven flows.
core/goals/__init__.py	Goal engine stub for future long-running autonomous goal execution.
core/goals/goal_engine.py	Goal engine stub for future long-running autonomous goal execution.
core/llm/__init__.py	LLM routing, response generation, reasoning prompt construction, and token tracking logic.
core/llm/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/__pycache__/llm_manager.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/__pycache__/provider_router.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/__pycache__/reasoning_engine.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/__pycache__/token_tracker.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/context_builder.py	LLM routing, response generation, reasoning prompt construction, and token tracking logic.
core/llm/llm_manager.py	LLM routing, response generation, reasoning prompt construction, and token tracking logic.
core/llm/provider_router.py	LLM routing, response generation, reasoning prompt construction, and token tracking logic.
core/llm/providers/__init__.py	LLM provider adapter implementing or stubbing provider-specific generation calls.
core/llm/providers/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/providers/__pycache__/anthropic_provider.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.

File	Purpose
core/llm/providers/__pycache__/base_provider.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/providers/__pycache__/groq_provider.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/providers/__pycache__/ollama_provider.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/providers/__pycache__/openai_provider.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/llm/providers/anthropic_provider.py	LLM provider adapter implementing or stubbing provider-specific generation calls.
core/llm/providers/base_provider.py	LLM provider adapter implementing or stubbing provider-specific generation calls.
core/llm/providers/groq_provider.py	LLM provider adapter implementing or stubbing provider-specific generation calls.
core/llm/providers/ollama_provider.py	LLM provider adapter implementing or stubbing provider-specific generation calls.
core/llm/providers/openai_provider.py	LLM provider adapter implementing or stubbing provider-specific generation calls.
core/llm/reasoning_engine.py	LLM routing, response generation, reasoning prompt construction, and token tracking logic.
core/llm/token_tracker.py	LLM routing, response generation, reasoning prompt construction, and token tracking logic.
core/memory/__init__.py	Memory manager or vector memory implementation backed by SQLite and ChromaDB.
core/memory/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/memory/__pycache__/memory_manager.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/memory/__pycache__/vector_memory.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/memory/advanced/long_term_knowledge.py	Placeholder or early-stage advanced memory component for future ranking, summarization, semantic learning, or organizational brain features.
core/memory/advanced/memory_ranker.py	Placeholder or early-stage advanced memory component for future ranking, summarization, semantic learning, or organizational brain features.
core/memory/advanced/memory_summarizer.py	Placeholder or early-stage advanced memory component for future ranking, summarization, semantic learning, or organizational brain features.
core/memory/advanced/organizational_brain.py	Placeholder or early-stage advanced memory component for future ranking, summarization, semantic learning, or organizational brain features.
core/memory/advanced/semantic_learning.py	Placeholder or early-stage advanced memory component for future ranking, summarization, semantic learning, or organizational brain features.
core/memory/memory_manager.py	Memory manager or vector memory implementation backed by SQLite and ChromaDB.
core/memory/vector_memory.py	Memory manager or vector memory implementation backed by SQLite and ChromaDB.
core/multimodal/document_parser.py	Placeholder multimodal component for document, screenshot, and vision routing features.

File	Purpose
<code>core/multimodal/multimodal_router.py</code>	Placeholder multimodal component for document, screenshot, and vision routing features.
<code>core/multimodal/screenshot_reasoner.py</code>	Placeholder multimodal component for document, screenshot, and vision routing features.
<code>core/multimodal/vision_engine.py</code>	Placeholder multimodal component for document, screenshot, and vision routing features.
<code>core/orchestration/__init__.py</code>	Core orchestration layer for agents, routing, workflows, collaboration, and system status.
<code>core/orchestration/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/orchestration/__pycache__/agent_registry.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/orchestration/__pycache__/collaboration_engine.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/orchestration/__pycache__/orchestration_manager.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/orchestration/__pycache__/task_router.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/orchestration/agent_registry.py</code>	Core orchestration layer for agents, routing, workflows, collaboration, and system status.
<code>core/orchestration/collaboration_engine.py</code>	Core orchestration layer for agents, routing, workflows, collaboration, and system status.
<code>core/orchestration/orchestration_manager.py</code>	Core orchestration layer for agents, routing, workflows, collaboration, and system status.
<code>core/orchestration/task_router.py</code>	Core orchestration layer for agents, routing, workflows, collaboration, and system status.
<code>core/planning/adaptive_planner.py</code>	Planning-related placeholder or helper component for task decomposition, dependencies, forecasting, and strategy.
<code>core/planning/dependency_graph.py</code>	Planning-related placeholder or helper component for task decomposition, dependencies, forecasting, and strategy.
<code>core/planning/execution_forecaster.py</code>	Planning-related placeholder or helper component for task decomposition, dependencies, forecasting, and strategy.
<code>core/planning/strategic_planner.py</code>	Planning-related placeholder or helper component for task decomposition, dependencies, forecasting, and strategy.
<code>core/planning/task_decomposer.py</code>	Planning-related placeholder or helper component for task decomposition, dependencies, forecasting, and strategy.
<code>core/reflection/reflection_engine.py</code>	Reflection/cross-agent critique engine using the LLM manager.
<code>core/state/__init__.py</code>	In-memory state manager for agent status, workflow status, system health, and checkpoints.
<code>core/state/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/state/__pycache__/state_manager.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>core/state/state_manager.py</code>	In-memory state manager for agent status, workflow status, system health, and checkpoints.
<code>core/workflows/__init__.py</code>	Workflow engine and predefined flagship workflow step definitions.
<code>core/workflows/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.

File	Purpose
core/workflows/__pycache__/flagship_workflows.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/workflows/__pycache__/workflow_engine.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
core/workflows/flagship_workflows.py	Workflow engine and predefined flagship workflow step definitions.
core/workflows/workflow_engine.py	Workflow engine and predefined flagship workflow step definitions.
dashboard/__init__.py	Python source module for project logic, scripts, tests, or utilities.
dashboard/static/css/app.css	CSS stylesheet for shared theme variables, base layout, landing page, dashboard, or app screens.
dashboard/static/css/base.css	CSS stylesheet for shared theme variables, base layout, landing page, dashboard, or app screens.
dashboard/static/css/dashboard.css	CSS stylesheet for shared theme variables, base layout, landing page, dashboard, or app screens.
dashboard/static/css/landing.css	CSS stylesheet for shared theme variables, base layout, landing page, dashboard, or app screens.
dashboard/static/css/theme.css	CSS stylesheet for shared theme variables, base layout, landing page, dashboard, or app screens.
dashboard/static/js/app.js	Browser-side JavaScript for auth, dashboard/app state, WebSocket updates, chat, onboarding, or hero animation.
dashboard/static/js/auth.js	Browser-side JavaScript for auth, dashboard/app state, WebSocket updates, chat, onboarding, or hero animation.
dashboard/static/js/dashboard.js	Browser-side JavaScript for auth, dashboard/app state, WebSocket updates, chat, onboarding, or hero animation.
dashboard/static/js/hero3d.js	Browser-side JavaScript for auth, dashboard/app state, WebSocket updates, chat, onboarding, or hero animation.
dashboard/templates/dashboard.html	HTML template served by FastAPI for a public page, auth page, dashboard, or logo.
dashboard/templates/landing.html	HTML template served by FastAPI for a public page, auth page, dashboard, or logo.
dashboard/templates/login.html	HTML template served by FastAPI for a public page, auth page, dashboard, or logo.
dashboard/templates/logo.html	HTML template served by FastAPI for a public page, auth page, dashboard, or logo.
dashboard/templates/signup.html	HTML template served by FastAPI for a public page, auth page, dashboard, or logo.
database/__init__.py	Python source module for project logic, scripts, tests, or utilities.
database/__pycache__/__init__.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
database/__pycache__/models.cpython-314.pyc	Python bytecode cache generated from a source module; not source of truth.
database/chroma_db/909af621-9653-4797-be07-b16f020b8c87/data_level0.bin	ChromaDB persistent vector-store storage file.
database/chroma_db/909af621-9653-4797-be07-b16f020b8c87/header.bin	ChromaDB persistent vector-store storage file.
database/chroma_db/909af621-9653-4797-be07-b16f020b8c87/length.bin	ChromaDB persistent vector-store storage file.

File	Purpose
database/chroma_db/909af621-9653-4797-be07-b16f020b8c87/link_lists.bin	ChromaDB persistent vector-store storage file.
database/chroma_db/chroma.sqlite3	ChromaDB persistent vector-store storage file.
database/mintuu_ecosystem.db	SQLite database file containing project runtime state and/or ChromaDB metadata.
database/models.py	SQLite schema and DatabaseManager CRUD/logging/statistics operations.
deployment/kubernetes/deployment.yaml	Kubernetes deployment manifest for the orchestrator service.
deployment/nginx/nginx.conf	Nginx reverse proxy/load-balancing configuration with WebSocket headers.
deployment/terraform/main.tf	Terraform AWS/EKS infrastructure sketch.
docker-compose.yml	Docker Compose definition for API, Celery worker, Redis, and Flower monitoring.
execution_trace.json	JSON trace/configuration artifact.
generate_architecture.py	Python source module for project logic, scripts, tests, or utilities.
integrations/__init__.py	Integration package marker; no concrete integration code currently lives here.
logs/workflow_1_execution.log	Execution log artifact from prior workflow runs.
mintuu_ai_ecosystem.egg-info/PKG-INFO	Project file or artifact used by the repository.
mintuu_ai_ecosystem.egg-info/SOURCES.txt	Plain text inventory, generated section, or raw reasoning log artifact.
mintuu_ai_ecosystem.egg-info/dependency_links.txt	Plain text inventory, generated section, or raw reasoning log artifact.
mintuu_ai_ecosystem.egg-info/entry_points.txt	Plain text inventory, generated section, or raw reasoning log artifact.
mintuu_ai_ecosystem.egg-info/requires.txt	Plain text inventory, generated section, or raw reasoning log artifact.
mintuu_ai_ecosystem.egg-info/top_level.txt	Plain text inventory, generated section, or raw reasoning log artifact.
mintuu_ecosystem.db	SQLite database file containing project runtime state and/or ChromaDB metadata.
mintuu_flagship_report.html	Generated or served HTML report/page artifact.
mintuu_master_doc.md	Markdown documentation or generated report artifact.
mintuu_master_doc.pdf	Generated PDF documentation or report artifact.
mintuu_master_doc_final.md	Markdown documentation or generated report artifact.
mintuu_master_doc_final.pdf	Generated PDF documentation or report artifact.
monitoring/orchestration_metrics.py	Monitoring/profiling/analytics/tracing module for system or workflow telemetry.
monitoring/performance_monitor.py	Monitoring/profiling/analytics/tracing module for system or workflow telemetry.
monitoring/system_profiler.py	Monitoring/profiling/analytics/tracing module for system or workflow telemetry.
monitoring/tracing_engine.py	Monitoring/profiling/analytics/tracing module for system or workflow telemetry.
monitoring/workflow_analytics.py	Monitoring/profiling/analytics/tracing module for system or workflow telemetry.

File	Purpose
<code>pyproject.toml</code>	Python package metadata, direct dependencies, optional dependency groups, entry point, and Ruff settings.
<code>raw_reasoning_log.txt</code>	Plain text inventory, generated section, or raw reasoning log artifact.
<code>render.yaml</code>	Render free-tier backend deployment configuration.
<code>report/.DS_Store</code>	macOS Finder metadata file; not part of application logic.
<code>report/output/mintuu_flagship_report.html</code>	Generated flagship report output artifact in HTML, Markdown, PDF, or image form.
<code>report/output/mintuu_flagship_report.md</code>	Generated flagship report output artifact in HTML, Markdown, PDF, or image form.
<code>report/output/mintuu_flagship_report.pdf</code>	Generated flagship report output artifact in HTML, Markdown, PDF, or image form.
<code>report/output/screenshots/chat_workflow.png</code>	Generated screenshot artifact from prior report/dashboard validation.
<code>report/output/screenshots/dashboard_bottom.png</code>	Generated screenshot artifact from prior report/dashboard validation.
<code>report/output/screenshots/dashboard_overview.png</code>	Generated screenshot artifact from prior report/dashboard validation.
<code>report/output/screenshots/dashboard_top.png</code>	Generated screenshot artifact from prior report/dashboard validation.
<code>report/output/screenshots/final_state.png</code>	Generated screenshot artifact from prior report/dashboard validation.
<code>report/output/screenshots/workflow_completed.png</code>	Generated screenshot artifact from prior report/dashboard validation.
<code>report/output/w1_dashboard.png</code>	Generated flagship report output artifact in HTML, Markdown, PDF, or image form.
<code>requirements_v3.txt</code>	Additional v3 AI, vector-memory, distributed-worker, and automation dependencies not all present in pyproject dependencies.
<code>run_flagships.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>run_flagships_v2.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>test_ollama.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>test_reasoning.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>test_workflow.py</code>	Python source module for project logic, scripts, tests, or utilities.
<code>tests/__init__.py</code>	Pytest package marker; no test modules are currently present.
<code>tools/__init__.py</code>	Tool registry or execution manager used by agents to access file, terminal, reporting, metrics, calendar, email, git, and utility tools.
<code>tools/__pycache__/__init__.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>tools/__pycache__/execution_manager.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.
<code>tools/__pycache__/tool_registry.cpython-314.pyc</code>	Python bytecode cache generated from a source module; not source of truth.

File	Purpose
<code>tools/execution_manager.py</code>	Tool registry or execution manager used by agents to access file, terminal, reporting, metrics, calendar, email, git, and utility tools.
<code>tools/implementations/__init__.py</code>	Standalone implementation for browser, email, or GitHub tool integration.
<code>tools/implementations/browser_tool.py</code>	Standalone implementation for browser, email, or GitHub tool integration.
<code>tools/implementations/email_tool.py</code>	Standalone implementation for browser, email, or GitHub tool integration.
<code>tools/implementations/github_tool.py</code>	Standalone implementation for browser, email, or GitHub tool integration.
<code>tools/tool_registry.py</code>	Tool registry or execution manager used by agents to access file, terminal, reporting, metrics, calendar, email, git, and utility tools.
<code>vercel.json</code>	Vercel frontend/static routing configuration that proxies API/auth/ws to the Render backend.

4. Current Working State

Area	Status	Evidence / Reason
FastAPI server	Working	Started successfully with <code>PYTHONPATH=/Users/mridulsoliwal/Documents/All_Projects/AI_Agents python3 -m uvicorn api.app:app --host 127.0.0.1 --port 8000</code> ; <code>/api/v1/status</code> returned healthy runtime state.
Landing page	Working	<code>GET /</code> returns <code>dashboard/templates/landing.html</code> .
Auth pages	Working	<code>GET /login</code> and <code>GET /signup</code> serve HTML; client-side <code>auth.js</code> calls <code>/auth/login</code> and <code>/auth/signup</code> .
Dashboard shell	Working with auth dependency	<code>GET /app</code> serves HTML, but browser JS redirects to <code>/login</code> if no local access token exists.
REST API	Working	OpenAPI exposes all core routes; <code>/api/v1/status</code> , <code>/api/v1/tools</code> , <code>/api/v1/agents</code> , <code>/api/v1/memory</code> , <code>/api/v1/analytics</code> were probed.
WebSocket endpoint	Working/partial	<code>/ws</code> accepts clients and pushes state every 2 seconds when clients are connected; client-to-server messages are only logged.
SQLite persistence	Working	<code>database/mintuu_ecosystem.db</code> has tables, indexes, and real rows: <code>{'agent_logs': 389, 'conversations': 18, 'events': 154, 'memories': 762, 'messages': 54, 'tasks': 326, 'tool_executions': 0, 'users': 1, 'workflow_states': 0, 'workflows': 15}</code> .
Agent registry	Working	Runtime has 9 active registered agents: CEO, HR, Marketing, Finance, Production, Operations, Research, Security, Infrastructure.
Analytics and Legal source agents	Partial/stubbed	<code>agents/analytics_agent/agent.py</code> and <code>agents/legal_agent/agent.py</code> exist but are not instantiated in <code>OrchestrationManager._init_agents</code> .
Product Strategy and Support agent folders	Stubbed/broken-empty	Folders exist with no source files.
Agent execution	Working with simulated domain logic	Agents return structured outputs from deterministic Python logic; <code>BaseAgent</code> wraps planning/execution/memory/logging.
LLM reasoning	Partial/simulated depending provider	Provider router chooses env provider; current <code>.env</code> has <code>ANTHROPIC_API_KEY</code> , so router selected Anthropic during startup, but <code>AnthropicProvider.generate()</code> returns mocked canned JSON and does not call the real API. Ollama/OpenAI/Groq paths are real adapters if configured.

Area	Status	Evidence / Reason
System prompts	Partial	CEO and Research use <code>SYSTEM_PROMPT</code> . Finance and Marketing define <code>FINANCE_SYSTEM_PROMPT</code> and <code>MARKETING_SYSTEM_PROMPT</code> , but dynamic import looks for <code>SYSTEM_PROMPT</code> , so those prompts are not actually used by BaseAgent without renaming or fallback logic.
ChromaDB vector memory	Working/partial	Persistent Chroma initializes; <code>org_knowledge</code> has 111 vectors. <code>workflows</code> , <code>decisions</code> , and <code>agent_memory</code> collections are created but empty.
Workflow engine	Working	15 completed workflows exist in SQLite; dynamic workflow engine supports dependencies, auto approval, state changes, retries, completion history.
Approval gates	Stubbed/simulated	<code>_handle_approval()</code> always returns True and auto-approves via CEO collaboration response.
Autonomous engine	Working but stopped by default	Default scheduled tasks exist; status reports <code>is_running: false</code> unless <code>/api/v1/autonomous/start</code> is called.
Tool registry	Working but security-light	11 tools registered; permissions default to EXECUTE, so READ/WRITE tools fail unless permissions are granted. Terminal tool uses <code>shell=True</code> , which is risky.
GitHub webhook	Partial	Endpoint accepts GitHub webhook JSON and triggers issue workflow, but no signature verification and event bus is an in-memory singleton.
Celery/Redis distributed layer	Partial/configured	Compose includes Redis/worker/Flower; Celery modules exist, but main API execution path is synchronous in-process.
Docker Compose	Partial/broken unless Dockerfile exists elsewhere	<code>docker-compose.yml</code> uses <code>build: .</code> , but no <code>Dockerfile</code> is present in this repo tree.
Render deployment	Partial	<code>render.yaml</code> exists; start command likely needs correct <code>PYTHONPATH</code> ; free tier has sleep/persistence limitations.
Vercel deployment	Partial	<code>vercel.json</code> rewrites static routes to <code>/templates/\$1.html</code> , but the actual root is <code>dashboard/templates</code> ; WebSocket rewrite uses HTTPS destination rather than WSS semantics.
Tests	Broken/incomplete	<code>python3 -m pytest tests/ -v</code> collected 0 tests and exited code 5. Test scripts exist at root but are not pytest modules in <code>tests/</code> .
Local import ergonomics	Partial	Running <code>python3 -m uvicorn api.app:app</code> from repo root failed until <code>PYTHONPATH=/Users/mridulsoaliwal/Documents/All_Projects/AI_Agents</code> was set.

5. All APIs and Endpoints

Method	URL	What it does	Parameters	Returns
GET	<code>/</code>	Serves landing page	none	HTML <code>landing.html</code>
GET	<code>/login</code>	Serves login page	none	HTML <code>login.html</code>
GET	<code>/signup</code>	Serves signup page	none	HTML <code>signup.html</code>
GET	<code>/app</code>	Serves authenticated dashboard shell	none server-side; client checks local token	HTML <code>dashboard.html</code>
POST	<code>/auth/signup</code>	Creates user and tokens	JSON: email, password, confirm_password, full_name	access token, refresh token, user
POST	<code>/auth/login</code>	Authenticates user	JSON: email, password	access token, refresh token, user
POST	<code>/auth/refresh</code>	Refreshes access token	Authorization: Bearer refresh_token	access_token, token_type
GET	<code>/auth/me</code>	Gets current profile	Authorization: Bearer access_token	profile object
PUT	<code>/auth/me</code>	Updates profile	Authorization plus JSON full_name/theme_preference	status updated
POST	<code>/auth/onboarding/complete</code>	Marks onboarding complete	Authorization	status

Method	URL	What it does	Parameters	Returns
POST	/auth/data/clear	Deletes user conversations and scoped memories	Authorization	status
GET	/auth/data/export	Exports user profile/conversations/workflows	Authorization	JSON export
POST	/api/v1/chat	Main chat endpoint; routes to task or workflow	JSON: message, optional conversation_id, user_id	conversation_id, response, result, timestamp
POST	/api/v1/execute	Executes a single task	JSON: description, optional agent_type, priority	agent task result
GET	/api/v1/tasks	Lists tasks from SQLite	query status, limit	array of task rows
POST	/api/v1/workflows	Creates and executes explicit workflow	JSON: name, description, steps[]	workflow result
POST	/api/v1/workflows/auto	Creates agent order from router and executes workflow	JSON: description	workflow result
GET	/api/v1/workflows	Lists active and recent in-memory workflows	none	array
GET	/api/v1/workflows/{workflow_id}	Gets active in-memory or DB workflow	path workflow_id	workflow object/row
GET	/api/v1/agents	Lists registered agents	none	array of agent info
GET	/api/v1/agents/{agent_type}	Gets one agent by type	path agent_type	agent info
GET	/api/v1/collaboration	Gets collaboration feed	query limit	active requests, history, stats, message history
GET	/api/v1/collaboration/messages	Gets message bus history	query limit	array of messages
GET	/api/v1/autonomous	Gets autonomous engine status	none	tasks and running state
GET	/api/v1/autonomous/events	Gets autonomous execution events	query limit	array
POST	/api/v1/autonomous/tasks	Registers scheduled task	JSON: name, agent_type, description, interval_seconds	task_id, status
POST	/api/v1/autonomous/{task_id}/trigger	Runs scheduled task immediately	path task_id	event object or 404
POST	/api/v1/autonomous/start	Starts recurring scheduler thread	none	status
POST	/api/v1/autonomous/stop	Stops recurring scheduler thread	none	status
GET	/api/v1/status	Full system status	none	ecosystem, agents, state, tools, memory, db stats, workflows, autonomous
GET	/api/v1/logs	Recent agent logs	query limit	array of agent_logs
GET	/api/v1/events	Recent system events	query limit	array of events
GET	/api/v1/tools	Lists registered tools	none	array of tool metadata
GET	/api/v1/memory	Memory cache stats	none	cache stats
GET	/api/v1/analytics	SQLite aggregate stats	none	counts by task/workflow status etc.

Method	URL	What it does	Parameters	Returns
POST	/webhooks/github	Receives GitHub webhook and triggers issue workflow	JSON GitHub payload plus X-GitHub-Event header	status accepted, event_id
WS	/ws	Dashboard push stream	WebSocket connection; client messages ignored except logging	JSON events: state_update, collaboration_update, autonomous_update

6. Database Schema

SQLite database path: database/mintuu_ecosystem.db.

agent_logs (389 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
agent_id	TEXT	True	None	False	Agent ID or pseudo-agent namespace such as organization/workflow.
action	TEXT	True	None	False	Agent action name.
input_data	TEXT	False	''	False	Serialized action input.
output_data	TEXT	False	''	False	Serialized action output.
status	TEXT	False	'SUCCESS'	False	Current lifecycle status.
duration_ms	INTEGER	False	0	False	Execution duration in milliseconds.
metadata	TEXT	False	'{'}	False	JSON metadata blob.
created_at	TEXT	True	None	False	UTC creation timestamp.

conversations (18 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
user_id	TEXT	True	'default'	False	Owning user identifier.
metadata	TEXT	False	'{'}	False	JSON metadata blob.
created_at	TEXT	True	None	False	UTC creation timestamp.
updated_at	TEXT	True	None	False	UTC update timestamp.

events (154 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
event_type	TEXT	True	None	False	System event type.
source	TEXT	True	None	False	Event source.
description	TEXT	True	None	False	Human-readable work description.
metadata	TEXT	False	'{'}	False	JSON metadata blob.
created_at	TEXT	True	None	False	UTC creation timestamp.

memories (762 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
agent_id	TEXT	True	None	False	Agent ID or pseudo-agent namespace such as organization/workflow.
memory_type	TEXT	True	None	False	Memory tier enum.
key	TEXT	True	None	False	Lookup key for memory item.

Column	Type	Required	Default	Primary Key	Meaning
content	TEXT	True	None	False	Message or memory content.
metadata	TEXT	False	'{'}	False	JSON metadata blob.
importance	REAL	False	0.5	False	Memory importance score.
access_count	INTEGER	False	0	False	Number of times memory was accessed.
created_at	TEXT	True	None	False	UTC creation timestamp.
updated_at	TEXT	True	None	False	UTC update timestamp.
expires_at	TEXT	False	None	False	Expiration timestamp for TTL memories.

messages (54 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
conversation_id	TEXT	True	None	False	Parent conversation ID.
role	TEXT	True	None	False	Message role, constrained to user/assistant/system/agent.
content	TEXT	True	None	False	Message or memory content.
metadata	TEXT	False	'{'}	False	JSON metadata blob.
created_at	TEXT	True	None	False	UTC creation timestamp.

tasks (326 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
title	TEXT	True	None	False	Task or document title.
description	TEXT	True	None	False	Human-readable work description.
status	TEXT	True	'PENDING'	False	Current lifecycle status.
assigned_agent	TEXT	True	None	False	Agent ID assigned to execute the task.
workflow_id	TEXT	False	None	False	Related workflow ID.
priority	INTEGER	False	5	False	Task priority from low to high.
dependencies	TEXT	False	'[]'	False	JSON list of task/step dependencies.
result	TEXT	False	None	False	Serialized execution result.
error_message	TEXT	False	None	False	Failure reason if task failed.
retry_count	INTEGER	False	0	False	Retry counter.
metadata	TEXT	False	'{'}	False	JSON metadata blob.
created_at	TEXT	True	None	False	UTC creation timestamp.
updated_at	TEXT	True	None	False	UTC update timestamp.
started_at	TEXT	False	None	False	UTC start timestamp.
completed_at	TEXT	False	None	False	UTC completion timestamp.

tool_executions (0 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
agent_id	TEXT	True	None	False	Agent ID or pseudo-agent namespace such as organization/workflow.
tool_name	TEXT	True	None	False	Executed tool name.
input_params	TEXT	False	'{'}	False	Serialized tool input parameters.
output_data	TEXT	False	''	False	Serialized action output.

Column	Type	Required	Default	Primary Key	Meaning
status	TEXT	False	'SUCCESS'	False	Current lifecycle status.
duration_ms	INTEGER	False	0	False	Execution duration in milliseconds.
error	TEXT	False	None	False	Tool execution error.
created_at	TEXT	True	None	False	UTC creation timestamp.

users (1 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
email	TEXT	True	None	False	User email.
password_hash	TEXT	True	None	False	bcrypt password hash.
full_name	TEXT	True	None	False	User display name.
is_verified	INTEGER	False	0	False	Boolean-ish integer verification flag.
onboarding_complete	INTEGER	False	0	False	Boolean-ish integer onboarding flag.
theme_preference	TEXT	False	'dark'	False	User UI theme preference.
created_at	TEXT	True	None	False	UTC creation timestamp.
updated_at	TEXT	True	None	False	UTC update timestamp.

workflow_states (0 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
workflow_id	TEXT	True	None	False	Related workflow ID.
step_index	INTEGER	True	None	False	Workflow step index for state snapshots.
state_data	TEXT	True	'{'	False	Serialized workflow state snapshot.
created_at	TEXT	True	None	False	UTC creation timestamp.

workflows (15 rows)

Column	Type	Required	Default	Primary Key	Meaning
id	TEXT	False	None	True	Unique row identifier.
name	TEXT	True	None	False	Workflow/user/tool name depending on table.
description	TEXT	True	None	False	Human-readable work description.
status	TEXT	True	'PENDING'	False	Current lifecycle status.
steps	TEXT	True	'['	False	Serialized workflow step definitions.
current_step	INTEGER	False	0	False	Current workflow step index.
total_steps	INTEGER	False	0	False	Total workflow step count.
result	TEXT	False	None	False	Serialized execution result.
initiated_by	TEXT	False	'mintuu'	False	User/system that initiated a workflow.
metadata	TEXT	False	'{'	False	JSON metadata blob.
created_at	TEXT	True	None	False	UTC creation timestamp.
updated_at	TEXT	True	None	False	UTC update timestamp.
started_at	TEXT	False	None	False	UTC start timestamp.
completed_at	TEXT	False	None	False	UTC completion timestamp.

Relationships

Relationship	Type	Defined by
messages.conversation_id -> conversations.id	Many messages per conversation	SQLite foreign key with cascade delete.
tasks.workflow_id -> workflows.id	Many tasks may belong to one workflow	SQLite foreign key with ON DELETE SET NULL .
workflow_states.workflow_id -> workflows.id	Many state snapshots per workflow	SQLite foreign key with cascade delete.
memories.agent_id	Logical relationship to agents, organization , or workflow:<id>	Not enforced by foreign key.
agent_logs.agent_id	Logical relationship to agents	Not enforced by foreign key.
tool_executions.agent_id	Logical relationship to agents	Not enforced by foreign key.

Current SQLite Contents

Table	Row count
agent_logs	389
conversations	18
events	154
memories	762
messages	54
tasks	326
tool_executions	0
users	1
workflow_states	0
workflows	15

Recent workflow samples:

ID	Name	Status	Current Step	Total Steps	Created At	Description
56290520-72d7-481b-82d6-997dfcde5f3f	Auto-Workflow: launh a new product with q3	COMPLETED	5	0	2026-05-27T19:13:23.464102+00:00	launh a new product with q3
57ac8a21-16a0-4857-a1a6-7a6a912b9515	Auto-Workflow: Launch a new product by Q3 with a \$50k budget	COMPLETED	5	0	2026-05-27T18:59:34.075147+00:00	Launch a new product by Q3 with a \$50k budget
079a8d35-626e-4e40-9b86-b06ecfffff6e	Auto-Workflow: Launch a new product by Q3 with a \$50k budget	COMPLETED	5	0	2026-05-24T10:46:44.938844+00:00	Launch a new product by Q3 with a \$50k budget
8c57d6ba-afad-4cf8-99cc-e9f440ee572a	Auto-Workflow: Run a system health check across all agents	COMPLETED	5	0	2026-05-22T07:45:43.712154+00:00	Run a system health check across all agents
0c459a6e-8e67-4b0f-9cfd-844fc4fdcf1c	Auto-Workflow: Run a system health check across all agents	COMPLETED	5	0	2026-05-22T07:07:08.163408+00:00	Run a system health check across all agents
3ce636be-291b-4f69-a8e1-ebf9aaed6392	Auto-Workflow: Run a comprehensive budget review	COMPLETED	0	0	2026-05-22T07:06:34.861038+00:00	Run a comprehensive budget review
05763404-05fa-49c8-b85a-72c1814e9bce	Auto-Workflow: Analyze our company	COMPLETED	2	0	2026-05-08T14:57:19.426731+00:00	Analyze our company KPIs and financial health

ID	Name	Status	Current Step	Total Steps	Created At	Description
	KPIs and financial health					
4f0cecdb-6e16-4313-ba7a-45cfff2fabcd	Auto-Workflow: Create a comprehensive company performance report	COMPLETED	1	0	2026-05-08T14:50:07.832357+00:00	Create a comprehensive company performance report
d0435f42-3eb4-40dd-aee1-2af3e88de399	Auto-Workflow: aunch a new product across all departments	COMPLETED	5	0	2026-05-08T14:47:59.219949+00:00	aunch a new product across all departments
6a27b555-b625-493f-91ac-7da303ea139d	Auto-Workflow: Now evolve Mintuu AI Ecosystem from a workflow dem	COMPLETED	4	0	2026-05-08T14:47:15.436869+00:00	Now evolve Mintuu AI Ecosystem from a workflow demo into a truly intelligent autonomous multi-agent

Recent task samples:

ID	Agent	Status	Title	Created At
fdccaa34-96d4-4012-8c0f-472614545abd	agent-research	COMPLETED	Task: analyze our competitor startegies	2026-05-27T19:14:43.317708+00:00
72a9eda8-ce69-404e-8238-8abe5742f78e	agent-finance	COMPLETED	Task: Review monthly burn rate and budget	2026-05-27T19:12:12.709566+00:00
e8d518d8-c52a-425c-9a9d-ccede01869c1	agent-production	COMPLETED	Task: Audit our infrastructure security policies	2026-05-24T10:49:17.705763+00:00
de2e5092-3575-41a4-8d6b-2461bf5ece24	agent-operations	COMPLETED	Task: Find the absolute path of logo.html in this project workspace by searching the f	2026-05-22T09:19:47.319316+00:00
0bec9315-ba8c-4e83-b6e9-34aa9ef094ac	agent-operations	COMPLETED	Task: Find the absolute path of logo.html in this project workspace by searching the f	2026-05-22T09:17:02.119676+00:00
6718ffe2-0f99-490b-a374-c57b427c0997	agent-operations	COMPLETED	Task: Test workflow	2026-05-22T09:08:38.943903+00:00
bd4214a2-3398-41a9-a54b-2b1fab711134	agent-research	COMPLETED	Task: Run competitor analysis and market watch	2026-05-22T07:06:57.105893+00:00
cca80d3d-7976-4fe9-b2dd-0ecf7b126c54	agent-operations	COMPLETED	Task: Monitor system health, workflow efficiency, and automation status	2026-05-09T07:17:44.464878+00:00
db6b21fd-3233-4bbe-813b-4f46fc3ee66d	agent-research	COMPLETED	Task: Analyze competitor activities, market trends, and industry news	2026-05-09T07:14:44.375260+00:00
6b993f5e-31a0-4699-9c52-1458bc074768	agent-operations	COMPLETED	Task: Monitor system health, workflow efficiency, and automation status	2026-05-09T07:14:44.369626+00:00

Recent memory samples:

ID	Agent	Type	Key	Content Preview
32c4ef60-5184-43f8-9227-b842e875ebf8	organization	ORGANIZATIONAL	task_result:fdccaa34-96d4-4012-8c0f-472614545abd	 Research Summary / Task: Task: analyze our competitor startegies / Confirmed historical precedent. Similar memory leak in data parser previously caused Se
7f3a341a-358a-46cb-b83d-0e94e8da04fd	organization	ORGANIZATIONAL	org_task:fdccaa34-96d4-4012-8c0f-472614545abd	 Research Summary / Task: Task: analyze our competitor startegies / Confirmed historical precedent. Similar memory leak in data parser previously caused Se

ID	Agent	Type	Key	Content Preview
be79b963-6946-4f6f-a4de-efb42114c2b3	agent-research	SHORT_TERM	task_result:fdccaa34-96d4-4012-8c0f-472614545abd	 Research Summary / Task: Task: analyze our competitor strategies / Confirmed historical precedent. Similar memory leak in data parser previously caused Se
dd98bc92-1fa0-4597-a0a8-50643222709b	workflow:56290520-72d7-481b-82d6-997dfcde5f3f	WORKFLOW	final_result	{ "0": "\ud83d\udcca CEO Executive Summary \nTask: CEO Analysis\nStatus: COMPLETED\n\n Decision: Task processed successfully.\n\n**\ud83e\udd16 LLM Thought
7660e7a4-0e63-4387-8a1f-3280e2af4ebe	organization	ORGANIZATIONAL	org_task:37870048-ae58-4c03-8008-29bad1bcc09b	 Operations Summary / Task: OPERATIONS Analysis / Response plan finalized: Isolate containers, revert commit f4a9b2, and deploy previous stable release.

| | 41346ef9-53cb-4552-a6db-489d2190f1a6 | agent-operations | SHORT_TERM | task_result:37870048-ae58-4c03-8008-29bad1bcc09b |  **Operations Summary** / Task: OPERATIONS Analysis / Response plan finalized: Isolate containers, revert commit f4a9b2, and deploy previous stable release.

| | a94f1858-8adc-4489-97c9-94cc3ebf6cd4 | organization | ORGANIZATIONAL | org_task:3937610b-7918-41f6-96c8-2efffca0a42f |  **Production Summary** / Task: PRODUCTION Analysis / Task processed successfully.

 **LLM Thought Process:** Analyzing the request. Proceeding with standard | | 85a9ae1e-734f-4985-b11c-1d4c7aa3c35c | agent-production | SHORT_TERM | task_result:3937610b-7918-41f6-96c8-2efffca0a42f |  **Production Summary** / Task: PRODUCTION Analysis / Task processed successfully.

 **LLM Thought Process:** Analyzing the request. Proceeding with standard | | 426faaf2-6015-442c-a9cb-e43e50b6f624 | organization | ORGANIZATIONAL | org_task:0ccb5939-914a-437c-80a3-bc18cf6b97e7 |  **Finance Summary** / Task: FINANCE Analysis / Task processed successfully.

 **LLM Thought Process:** Analyzing the request. Proceeding with standard execut | | 89436dfc-e21d-4127-ba6a-e325419cb19d | agent-finance | SHORT_TERM | task_result:0ccb5939-914a-437c-80a3-bc18cf6b97e7 |  **Finance Summary** / Task: FINANCE Analysis / Task processed successfully.

 **LLM Thought Process:** Analyzing the request. Proceeding with standard execut |

ChromaDB Collections

Persistent path: database/chroma_db.

Collection	Count	Purpose
decisions	0	Collection is created but currently empty/unused
agent_memory	0	Collection is created but currently empty/unused
workflows	0	Collection is created but currently empty/unused
org_knowledge	111	Stores shared task/workflow result embeddings

7. Agent System Details

Runtime Registered Agents

Agent	ID	Responsibility	LLM model preference / prompt state	Outputs
CEO	agent-ceo	Strategic leadership, approvals, KPI analysis, goal management	llama3.1 preference through reasoning engine; provider selected by router	Strategic plans, approvals, executive summaries
HR	agent-hr	Hiring, onboarding, employee/team workflows, meetings	mistral preference through default BaseAgent path	HR reports, candidate/onboarding/team outputs
Marketing	agent-marketing	Campaigns, SEO, content, growth strategy	mistral preference; CMO prompt exists but dynamic import currently looks for	Marketing strategy and campaign summaries

Agent	ID	Responsibility	LLM model preference / prompt state	Outputs
			<code>SYSTEM_PROMPT</code> , so default prompt is used unless fixed	
Finance	<code>agent-finance</code>	Budgets, expenses, revenue, forecasts	mistral preference; CFO prompt exists as <code>FINANCE_SYSTEM_PROMPT</code> but dynamic import expects <code>SYSTEM_PROMPT</code> , so default prompt is used unless fixed	Budget/financial summaries
Production	<code>agent-production</code>	Deployments, release, CI/CD simulation, health monitoring	mistral preference	Deployment and health summaries
Operations	<code>agent-operations</code>	Workflow coordination, automation scheduling, status reporting	llama3.1 preference	Operational plans and status summaries
Research	<code>agent-research</code>	Market research, competitor analysis, vector memory lookup	llama3.1 preference and Research system prompt	Research summaries with vector-memory context
Security	<code>agent-security</code>	Risk assessment, threat modeling, incident response	mistral preference	Risk rating and security summary
Infrastructure	<code>agent-infrastructure</code>	Infrastructure health/status/scaling	mistral preference	Infrastructure status and blast radius
Analytics	<code>agent-analytics</code>	Data processing/insights; source exists but not registered by orchestrator	not active	Basic metrics output if manually instantiated
Legal	<code>agent-legal</code>	Compliance/contract/risk; source exists but not registered by orchestrator	not active	Compliance recommendation if manually instantiated

All runtime agents inherit `BaseAgent`, which provides `handle_task()`: set state to planning, call `plan()`, optionally call `ReasoningEngine.reason()`, call domain `execute()`, merge LLM final decision into outputs, summarize, broadcast reasoning trace, log activity, save short-term memory, save organizational vector memory, update state to idle, and return a result record.

Tools Available to Agents

Runtime `/api/v1/tools` verified 11 tools: `file_read`, `file_write`, `terminal_execute`, `data_analysis`, `report_generator`, `git_status`, `system_info`, `document_generator`, `calendar_manager`, `email_send`, `metrics_collector`.

Permissions are present but underdeveloped. Default agent permissions are `[EXECUTE]`, so tools requiring `READ` or `WRITE` are blocked unless `ToolRegistry.grant_permission()` is called.

System Prompts

CEO prompt in `agents/ceo_agent/system_prompt.py`:

```
You are the Chief Executive Officer (CEO) of Mintuu AI.
Your role is to make high-level strategic decisions, review plans, and guide the overall direction of the company.

CRITICAL INSTRUCTION:
1. You MUST read the provided CONTEXT (Research/Production/Marketing outputs).
2. Your "thought_process" and "final_decision" MUST explicitly reference specific metrics, file names, risks, or past incidents.
3. FINANCIAL RIGOR: You are hyper-critical of marketing and business plans.
   - If a plan assumes a conversion rate > 3% for new developer signups without a massive proven community, you MUST reject it as "mathematically optimistic".
   - You must require a REVISION if the numbers don't add up to the stated goals (e.g. 500 signups / 50k revenue).
   - If you reject a plan, start your final_decision with "REJECTED: [Reason]".
   - If you approve, start with "APPROVED: [Rationale]".

Be concise, authoritative, and strategically demanding.
```

Research prompt in `agents/research_agent/system_prompt.py`:

You are the Research Agent of Mintuu AI.
Your role is to conduct market research, competitor analysis, and gather information from past incidents using vector memory.

CRITICAL INSTRUCTION:
You MUST read the provided CONTEXT, which will include `vector\_memory\_results`. Your "thought_process" and "final_decision" MUST explicitly reference the specific details from these vector memory results.
DO NOT invent generic findings. Extract the actual past incidents, metrics, and resolutions from the memory and present them specifically in your final decision.

Finance and Marketing prompts exist but are not automatically loaded because their variables are named `FINANCE_SYSTEM_PROMPT` and `MARKETING_SYSTEM_PROMPT`, while `BaseAgent` imports `SYSTEM_PROMPT`.

8. Workflow Definitions

Dynamic Auto Workflow

Trigger: `/api/v1/chat` when message contains workflow keywords (`launch`, `full report`, `company`, `all departments`, `comprehensive`, `complete analysis`, `across all`, `new product`) or direct `/api/v1/workflows/auto`.

Router logic: `TaskRouter.suggest_workflow()` returns `['ceo', 'research', 'marketing', 'finance', 'production', 'operations']` for company-wide/new-product signals; otherwise it ranks agents by keyword rules and keeps CEO first and Operations last.

Step format:

Field	Meaning
<code>agent_type</code>	Agent type to run.
<code>title</code>	Step title.
<code>description</code>	Department-specific task description.
<code>depends_on</code>	Previous step index by default.
<code>requires_approval</code>	True for first CEO step when CEO is in the workflow; approval is currently auto-approved.

Flagship Workflow: GitHub Incident Pipeline

Defined in `core/workflows/flagship_workflows.py`, triggered by `POST /webhooks/github` when `X-GitHub-Event: issues` and payload action is `opened`.

Step	Agent	Title	Dependencies	Final output contribution
1	Research	Analyze Issue against Memory	none	Past incident and resolution report from vector memory.
2	Production	Impact Analysis	step 1	Affected files/recent commit impact report.
3	CEO	Severity Decision	steps 1 and 2	Severity and immediate action decision.
4	Operations	Concrete Response Plan	step 3	Concrete response plan.
5	Marketing	Final Summary & Update	step 4	Final public/update summary and labels.

Flagship Workflow: Product Launch

Defined by `get_product_launch_steps(goal)`.

Step	Agent	Title	Dependencies	Final output contribution
1	CEO	Decompose Goal	none	Phases, budget range, success metrics.
2	Research	Market Retrieval	step 1	Market/competitor/pricing retrieval.
3	Marketing	Campaign Plan	step 2	Campaign plan based on research.
4	CEO	Plan Critique Loop	step 3	Critique against original goal metrics.
5	Finance	Allocate Budget	step 4	Budget allocation tied to constraints.
6	Operations	Execution Checklist & KPIs	step 5	Sequenced execution checklist and KPI checkpoints.

Flagship Workflow: Incident Response

Defined by `get_incident_response_steps(anomaly)`.

Step	Agent	Title	Dependencies	Final output contribution
1	Infrastructure	Status Report	none	Factual affected-services and blast-radius report.
2	Security	Risk Assessment	step 1	Exploit/risk rating.
3	Operations	Severity Scoring	steps 1 and 2	Formula-based severity score and escalation need.
4	CEO	Escalation Decision	step 3	Approval decision and conditions.
5	Operations	Post-Mortem Generation	step 4	Plain-English post-mortem.

Approval Gates

Workflow steps support `requires_approval`. In current code, `_handle_approval()` creates a collaboration approval request to `agent-ceo`, auto-responds with approved data, logs an approval gate, and always returns `True`. This is simulation, not real human approval.

9. Memory System Details

Memory is implemented in `core/memory/memory_manager.py`, `core/memory/vector_memory.py`, and `database/models.py`.

End-to-end flow:

- 1. A task is executed by `OrchestrationManager.execute_task()` or `WorkflowEngine._execute_step()`.
- 2. Agent context is built by `MemoryManager.build_agent_context()` from short-term memory, long-term memory, optional conversation messages, optional workflow memory, and organizational memory.
- 3. `ResearchAgent.handle_task()` additionally queries `MemoryManager.query_vector_memory()` against Chroma collection `org_knowledge` and injects `vector_memory_results` into context.
- 4. `BaseAgent.handle_task()` executes the agent, summarizes output, logs the activity, saves short-term memory, and saves organizational memory.
- 5. `MemoryManager.store_organizational()` stores in SQLite `memories` and also writes an embedding document to Chroma `org_knowledge`.
- 6. Completed workflow final results are stored as workflow memory with agent ID `workflow:<workflow_id>` and key `final_result`.

Memory tiers:

Tier	Stored in SQLite type	Agent namespace	TTL	Vectorized
Short-term	SHORT_TERM	concrete agent ID	1 hour default	No
Long-term	LONG_TERM	concrete agent ID	none	No
Workflow	WORKFLOW	workflow:<id>	72 hours	No
Conversation	CONVERSATION	mintuu	none	No
Organizational	ORGANIZATIONAL	organization	none	Yes, in <code>org_knowledge</code>

Current memory distribution top rows:

Memory Type	Agent Namespace	Count
ORGANIZATIONAL	organization	362
SHORT_TERM	agent-operations	111
SHORT_TERM	agent-finance	82
SHORT_TERM	agent-marketing	68
SHORT_TERM	agent-ceo	63
SHORT_TERM	agent-research	55
SHORT_TERM	agent-production	10
WORKFLOW	workflow:05763404-05fa-49c8-b85a-72c1814e9bce	1
WORKFLOW	workflow:079a8d35-626e-4e40-9b86-b06ecfffff6e	1
WORKFLOW	workflow:0c459a6e-8e67-4b0f-9cfd-844fc4fdcf1c	1
WORKFLOW	workflow:3ce636be-291b-4f69-a8e1-ebf9aaed6392	1
WORKFLOW	workflow:4f0cecd6-6e16-4313-ba7a-45cffffc2fab	1

Memory Type	Agent Namespace	Count
WORKFLOW	workflow:56290520-72d7-481b-82d6-997dfcde5f3f	1
WORKFLOW	workflow:57ac8a21-16a0-4857-a1a6-7a6a912b9515	1
WORKFLOW	workflow:6a27b555-b625-493f-91ac-7da303ea139d	1
WORKFLOW	workflow:8c57d6ba-afad-4cf8-99cc-e9f440ee572a	1
WORKFLOW	workflow:8d60bee7-13dd-4de3-aa8c-ba73fa26bbad	1
WORKFLOW	workflow:d0435f42-3eb4-40dd-ae1-2af3e88de399	1

10. Frontend Details

HTML Pages

Template	URL	Purpose
dashboard/templates/landing.html	/	Public marketing/landing page with nav, hero, demo, comparison, CTA, footer, hero animation.
dashboard/templates/login.html	/login	Login form wired to <code>handleLogin()</code> in <code>auth.js</code> .
dashboard/templates/signup.html	/signup	Signup form wired to <code>handleSignup()</code> in <code>auth.js</code> .
dashboard/templates/dashboard.html	/app	Main authenticated dashboard shell using Alpine.js controller from <code>app.js</code> .
dashboard/templates/logo.html	not directly routed	Standalone logo/design artifact.

JavaScript Files

File	Purpose
dashboard/static/js/auth.js	Stores tokens in localStorage, signs up/logs in, refreshes tokens, wraps authenticated fetches, redirects authenticated users away from auth pages, and logs out.
dashboard/static/js/app.js	Main Alpine dashboard controller: auth gate, onboarding, tour, chat, state fetch, WebSocket connection, page navigation, workflow/agent data state, toasts, settings/history helpers.
dashboard/static/js/dashboard.js	Older/alternate dashboard controller: polling, WebSocket state handling, collaboration rendering, workflow graph/timeline rendering, logs, chat.
dashboard/static/js/hero3d.js	Landing hero Three.js particle network with mobile/reduced-motion SVG fallback, FPS degradation, mouse parallax, pulse edges.

CSS Files

File	Purpose
dashboard/static/css/theme.css	Theme tokens/colors/design variables.
dashboard/static/css/base.css	Shared reset/base layout and reusable components.
dashboard/static/css/landing.css	Public landing page layout, hero, demo, sections, CTA, footer.
dashboard/static/css/app.css	Authenticated app/dashboard layout and components.
dashboard/static/css/dashboard.css	Older/alternate dashboard styling.

Current UI state: the app has a polished dark AI dashboard aesthetic, onboarding overlay, guided tour spotlight, chat panel, agents/workflows/history/settings navigation, real-time status labels, toasts, and a landing hero animation. It still needs cleanup because `app.js` and `dashboard.js` overlap; dashboard auth is client-side only; some UI data depends on in-memory workflow history, so completed DB workflows do not all appear after restart.

Animations/interactions include rotating chat placeholders, welcome animation, guided tour spotlight/tooltip, toasts, page transitions, WebSocket-driven state refresh, Three.js node/edge animation, mobile/reduced-motion hero fallback, FPS-based hero degradation, and dashboard polling fallback.

11. Authentication System

Auth files: `api/auth.py`, `api/auth_routes.py`, `dashboard/static/js/auth.js`.

Aspect	Implementation
Signup	<code>/auth/signup</code> validates matching passwords and minimum length 6, creates a SQLite user, returns tokens.
Login	<code>/auth/login</code> verifies bcrypt hash and returns tokens.
Password storage	Direct <code>bcrypt.hashpw()</code> with <code>gensalt(rounds=12)</code> .
Access token	JWT signed with <code>settings.auth.secret_key</code> , algorithm <code>HS256</code> , valid for 7 days. Payload: <code>sub, email, name, exp, iat, type: access</code> .
Refresh token	JWT signed with same secret and algorithm, valid for 30 days. Payload: <code>sub, exp, iat, type: refresh</code> .
Client storage	<code>localStorage</code> keys: <code>mintuu_access_token</code> , <code>mintuu_refresh_token</code> , <code>mintuu_user</code> .
Protected routes server-side	<code>/auth/me</code> , <code>/auth/me PUT</code> , <code>/auth/onboarding/complete</code> , <code>/auth/data/clear</code> , <code>/auth/data/export</code> .
Public API routes	All <code>/api/v1/*</code> routes, <code>/webhooks/github</code> , page routes, docs are public at server level.
Dashboard protection	Client-side only; <code>/app</code> HTML is public but JS redirects if no token exists.
Risk	<code>SECRET_KEY</code> defaults to a random process-local value if not configured, invalidating tokens after restart and making multi-instance deployment inconsistent.

12. LLM Integration Details

Provider router logic in `core/llm/provider_router.py`:

1. If `DEFAULT_LLM_PROVIDER` is set and matches a provider, use it.
2. Else if `GROQ_API_KEY` exists, use Groq.
3. Else if `OPENAI_API_KEY` exists, use OpenAI.
4. Else if `ANTHROPIC_API_KEY` exists, use Anthropic.
5. Else use Ollama.

Current local `.env` key names detected: `ANTHROPIC_API_KEY=<redacted>` and `FISH_API_KEY=<redacted>`. Because Anthropic key exists and no Groq/OpenAI key was detected, startup selected `anthropic`.

Provider behavior:

Provider	File	Model	Current behavior
OpenAI	<code>core/llm/providers/openai_provider.py</code>	<code>OPENAI_MODEL</code> or <code>gpt-4o-mini</code>	Real async OpenAI Chat Completions call.
Ollama	<code>core/llm/providers/ollama_provider.py</code>	<code>OLLAMA_MODEL</code> , mapped <code>llama3.1 -> llama3:latest</code> , <code>mistral -> mistral:latest</code>	Real local HTTP call to <code>/api/chat</code> after <code>/api/tags</code> health check.
Anthropic	<code>core/llm/providers/anthropic_provider.py</code>	<code>claude-3-haiku-20240307</code>	Mocked. It sleeps 2 seconds and returns canned JSON based on system prompt.
Groq	<code>core/llm/providers/groq_provider.py</code>	<code>GROQ_MODEL</code> or <code>llama-3.1-8b-instant</code>	Real OpenAI-compatible Groq HTTP call with optional JSON response format.

Agent model preference in `BaseAgent.handle_task()`:

```
pref = "llama3.1" if self.agent_type.lower() in ["ceo", "research", "operations"] else "mistral"
```

Important nuance: `model_preference` is also passed to `ProviderRouter.get_provider()`, which treats it as a provider name first. Because `llama3.1` and `mistral` are not provider names, the default provider is used, and the model preference only matters inside providers that inspect it, especially Ollama and Groq.

Reasoning output schema requested from LLM:

```
{
  "thought_process": "Detailed step-by-step reasoning",
  "required_tools": ["tool1", "tool2"],
  "action_plan": [{"step": 1, "action": "...", "expected_outcome": "..."}],
  "confidence_score": 0.95,
```

```
"final_decision": "Summary of what to do"
}
```

Parsing: `ReasoningEngine.reason()` strips markdown JSON fences if present, parses JSON, warns on confidence below 0.6, and returns fallback object with `raw` if JSON parsing fails.

13. Deployment Configuration

Environment Variables

Variable	Purpose	Required?
MINTUU_DATA_DIR	Data/database root; defaults to repo <code>database</code> .	No locally, yes for stable deploy storage.
LOG_LEVEL	Logging verbosity.	No.
PORT / API_PORT	API port; defaults to 8003 in settings.	Required by Render via <code>\$PORT</code> .
SECRET_KEY	JWT signing key.	Strongly required in production.
ENVIRONMENT	<code>development</code> / <code>production</code> label.	No.
ALLOWED_ORIGINS	Comma-separated CORS origins.	Required for deployed frontend.
DEFAULT_LLM_PROVIDER	Explicit provider selector.	No.
OLLAMA_BASE_URL	Local Ollama API base.	Needed for Ollama.
OLLAMA_MODEL	Default Ollama model.	No.
OPENAI_API_KEY	OpenAI provider credential.	Needed for OpenAI.
ANTHROPIC_API_KEY	Anthropic provider credential.	Needed only if real Anthropic implementation is restored.
GROQ_API_KEY	Groq provider credential; preferred for free Render deployment.	Needed for Groq.
GROQ_MODEL	Groq model override.	No.
REDIS_URL	Redis/Celery broker URL.	Needed for distributed worker.
CHROMADB_PATH	Configured Chroma path but current <code>VectorMemory()</code> uses its own default unless passed.	No/currently not fully wired.
GITHUB_TOKEN	GitHub tool token; missing means GitHub tool is simulated/limited.	Optional.
MINTUU_FRONTEND_URL	Frontend URL for CORS/email links.	Optional.

Local Run Commands

Known working command from this session:

```
cd /Users/mridulsoliwal/Documents/All_Projects/AI_Agents/mintuu_ai_ecosystem
PYTHONPATH=/Users/mridulsoliwal/Documents/All_Projects/AI_Agents python3 -m uvicorn api.app:app --host 127.0.0.1 --port 8000
```

Makefile command:

```
make dev
# expands to: PYTHONPATH=$(dirname $(pwd)) uvicorn api.app:app --port 8003 --reload
```

README command:

```
docker-compose up -d
python3 -m uvicorn api.app:app --host 0.0.0.0 --port 8000
```

Known deployment issues on free tier:

Platform	Issue
Render free	Service sleeps, SQLite and Chroma persistence may be unstable unless persistent disk is configured, background threads stop during sleep, local Ollama unavailable, <code>SECRET_KEY</code> must be stable, and <code>PYTHONPATH</code> /package layout must be correct.
Vercel	Current config tries to route templates as static files under <code>/templates</code> , but repo structure is <code>dashboard/templates</code> ; Vercel is not serving the FastAPI process.
Docker Compose	No Dockerfile exists, so <code>build: .</code> will fail unless a Dockerfile is added.
Kubernetes/Terraform	Manifests are sketches; missing service, ingress, secrets, Redis, persistent storage, Docker image build/push, IAM role definitions, subnets, and outputs.

14. Known Issues and Limitations

Issue	Why it exists	Fix
Version naming drift	README, app title, settings, and package metadata were updated independently.	Pick one version and update <code>pyproject.toml</code> , README, FastAPI title, logs, and settings.
Import requires parent <code>PYTHONPATH</code>	Code imports <code>mintuu_ai_ecosystem.*</code> while uvicorn target is <code>api.app:app</code> from inside package folder.	Run as installed package from parent, adjust package layout, or use <code>PYTHONPATH</code> consistently.
No tests collected	<code>tests/</code> only contains <code>__init__.py</code> ; root scripts are not pytest tests.	Add pytest files under <code>tests/</code> for API, DB, workflow, memory, auth.
Anthropic provider is mocked	Provider returns canned JSON with comment <code>MOCK ANTHROPIC SINCE CREDITS ARE 0.</code>	Implement real Anthropic call or rename provider to mock in dev.
Marketing/Finance prompts not loaded	Variable names do not match BaseAgent dynamic import expectation.	Rename to <code>SYSTEM_PROMPT</code> or update BaseAgent to detect agent-specific prompt variable names.
Approval gates are fake	<code>_handle_approval()</code> always approves.	Add persistent approval requests, dashboard UI, protected approve/reject endpoints.
Tool permissions are incomplete	Defaults allow EXECUTE only; no robust per-agent policy setup.	Define default role permissions and enforce sandbox paths/command allowlists.
Terminal tool is dangerous	Uses <code>subprocess.run(..., shell=True)</code> .	Replace with argument-list execution, allowlist commands, no shell by default, timeouts, approvals, and audit logs.
File tools are unsandboxed	<code>file_read</code> and <code>file_write</code> accept arbitrary paths.	Restrict to configured workspace roots and block secrets/system paths.
GitHub webhook has no signature verification	Endpoint accepts arbitrary JSON.	Verify <code>X-Hub-Signature-256</code> with webhook secret.
Dashboard auth is client-side	<code>/app</code> route always serves HTML; JS redirects.	Add server-side auth or accept this for SPA but protect APIs.
Core APIs are public	<code>/api/v1/*</code> endpoints do not require JWT.	Add auth dependencies and user scoping for production.
Chroma path setting not fully wired	<code>MemoryManager</code> instantiates <code>VectorMemory()</code> without passing <code>settings.CHROMADB_PATH</code> .	Pass configured path to <code>VectorMemory(settings.CHROMADB_PATH)</code> .
Duplicate/old dashboard controllers	<code>app.js</code> and <code>dashboard.js</code> overlap.	Consolidate into one dashboard controller.
Docker Compose likely fails	<code>docker-compose.yml</code> references <code>build: .</code> but no Dockerfile found.	Add Dockerfile and correct worker import path/PYTHONPATH.
Celery not central to execution	Main flow executes synchronously in API process.	Move workflow/agent execution to Celery or remove distributed claims.
Some feature modules are placeholders	adaptation, planning, multimodal, advanced memory, approval components are skeletal.	Either implement or mark roadmap modules clearly.
Generated artifacts are committed with source	PDFs, PNGs, DBs, pyc, reports, <code>.DS_Store</code> exist in tree.	Add <code>.gitignore</code> , move artifacts to <code>artifacts/</code> , avoid committing DB/pycache.
SQLite user data deletion incomplete	<code>delete_user_data()</code> deletes conversations but not tasks/workflows by user comprehensively.	Add user scoping to tasks/workflows/memories and transactional cleanup.

15. What Has Been Proven Working

Capability	Proof
Server startup	Uvicorn started on <code>http://127.0.0.1:8000</code> ; logs showed database, memory, state, message bus, tools, LLM manager, 9 agents, workflow engine, autonomous engine initialized.
Status API	<code>/api/v1/status</code> returned ecosystem name <code>Mintuu AI Ecosystem</code> , version <code>3.0.0</code> , environment <code>development</code> , 9 agents, 11 tools, and healthy state.
Tools API	<code>/api/v1/tools</code> returned all 11 tool metadata entries.
Agents API	<code>/api/v1/agents</code> returned 9 active runtime agents with states and capabilities.
Memory API	<code>/api/v1/memory</code> returned cache stats: cache capacity 500 and active cache entries.
Analytics API	<code>/api/v1/analytics</code> returned counts: 326 completed tasks, 15 completed workflows, 762 memories, no failed tasks/workflows.
SQLite schema	<code>.schema</code> inspection showed conversations, messages, tasks, workflows, workflow_states, memories, agent_logs, events, tool_executions, users, and indexes.
Existing workflow history	SQLite has 15 completed workflows including product launch, system health, budget review, KPI/financial health, and company performance report workflows.
Existing task history	SQLite has 326 completed tasks and 0 failed tasks.
Existing memory history	SQLite has 762 memories; Chroma <code>org_knowledge</code> has 111 embedded documents.
Frontend pages	<code>GET /, /app, /docs</code> returned HTML during verification.
Tests status	<code>python3 -m pytest tests/ -v</code> ran successfully as a command but collected 0 tests and exited code 5, proving test suite is missing rather than failing assertions.

Strategic Direction Notes for External AI

The best next architectural direction is to stabilize the core first: unify versions, fix import/package layout, protect APIs, replace mocked LLM/provider behavior with explicit dev/production modes, enforce tool sandboxing, add real tests, and separate source code from generated artifacts. After that, integrate `AI_Agents/Mac_Agent` as a separate local automation service exposed to Mintuu through a strict tool adapter and approval layer, rather than merging unsafe file/terminal capabilities directly into agent code.